LiDAR Dectection System

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Project Exposition

- Create a working sensor that can detect any changes in the movement/position of any nearby object.
- System rotates 180 degree and back to measure any changes in distance.

Construction

- Arduino Uno
- Small LIDAR range finding sensor (detect changes from a distance of up to 170 cm)
- Micro-servo motor
- Solderless breadboard
- 5V battery

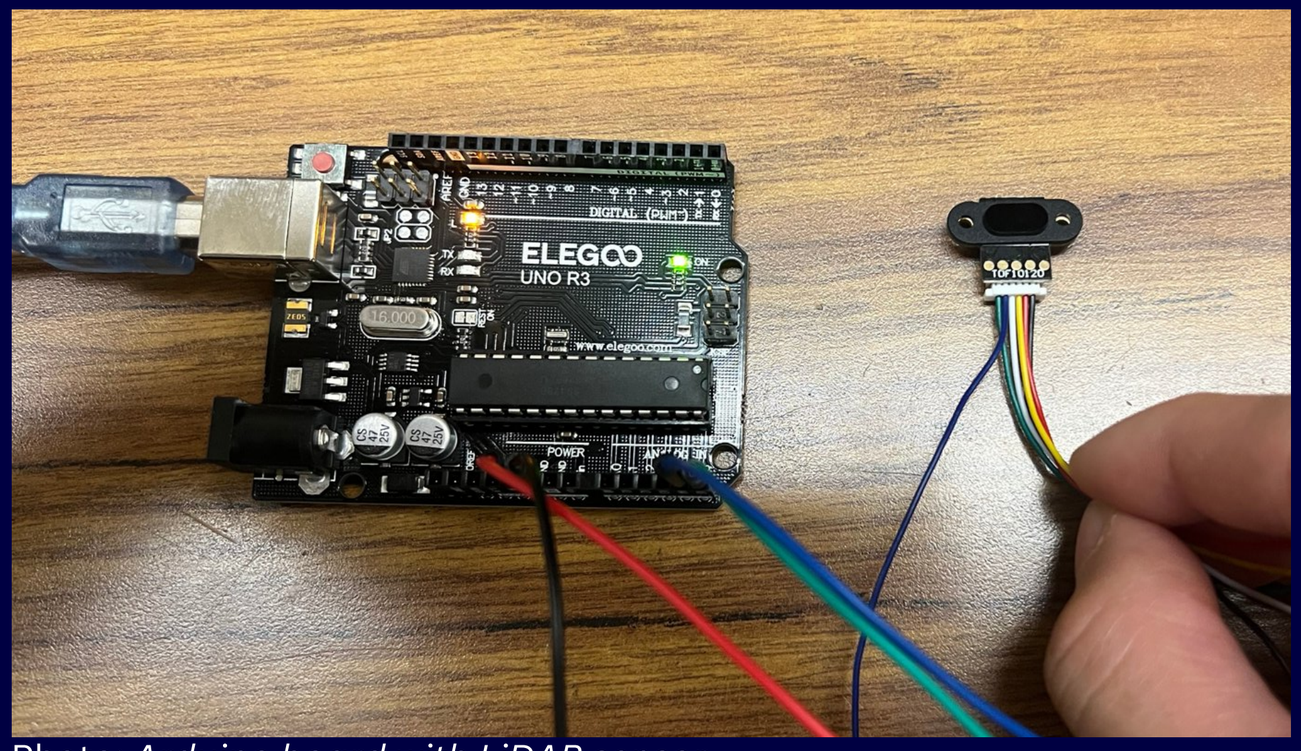
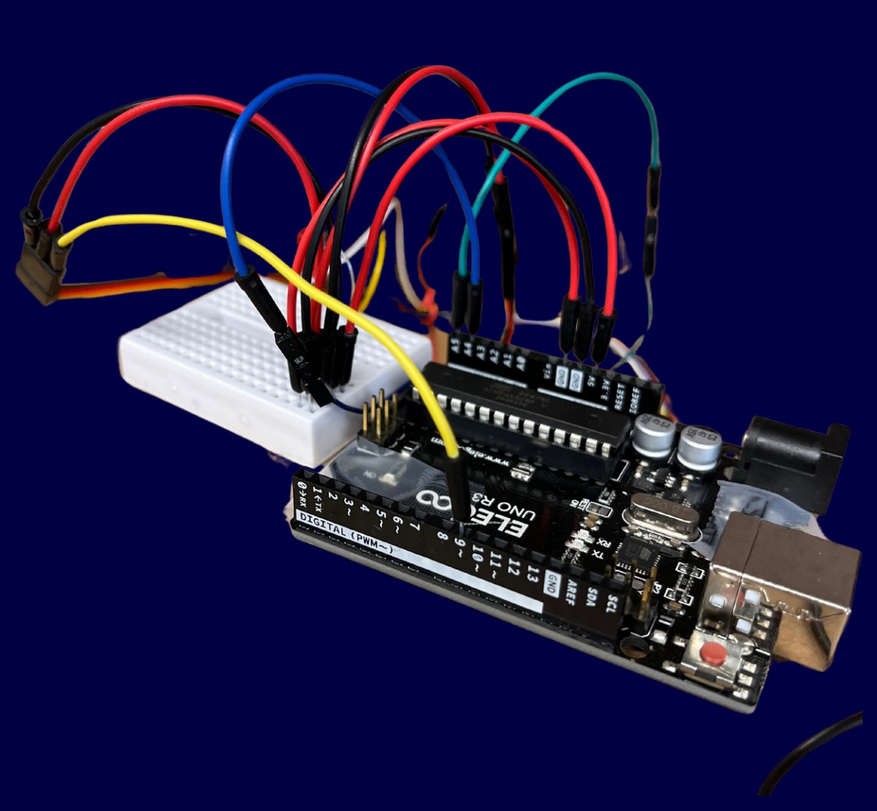


Photo: *Arduino board with LiDAR sensor*



Functionality

- Functions were developed to enable LiDAR to detect changes in object distance or disappearance from sight.
- Code was defined to initialize pins, servo start/stop angles, delays, and distance thresholds for quick system response.
- Servo speed was reduced from its maximum to provide meaningful data output.
- LED lights up only when pointing directly at the direction of detected change.

```
void loop() {
  sweepServo(0, 180, delta);  // Forward sweep
  sweepServo(180, 0, -delta); // Backward sweep
}

void sweepServo(int startAngle, int endAngle, int step) {
  int count = (startAngle == 0) ? 0 : 180 / delta;

  for (int angle = startAngle; angle != endAngle + step; angle += step)
{   myservo.write(angle);
    delay(30); // Allow the servo to stabilize
    int x = ReadDistance();

    // Check for significant change
    if (abs(x - position[count]) > threshold) {
      ledttl += 2; // Increase LED warning duration
    }
    if (ledttl > 0) ledttl--;

    // Update position array
    position[count] = x;

    // Control LED based on `ledttl`
    digitalWrite(12, (ledttl > 0) ? HIGH : LOW);

    // Debug output
    Serial.print("Angle: ");
    Serial.print(angle);
    Serial.print(" Distance: ");
    Serial.print(x);
    Serial.print(" mm, LED TTL: ");
    Serial.println(ledttl);

    // Update counter
    count += (step > 0) ? 1 : -1;
  }
}
```

```
void SensorRead(unsigned char addr, unsigned char *datbuf, unsigned char cnt)
{ Wire.beginTransmission(sensorAddress);
  Wire.write(byte(addr));
  Wire.endTransmission();
  delay(1); // Allow sensor time to process
  Wire.requestFrom(sensorAddress, cnt);

  for (int i = 0; i < cnt && Wire.available(); i++) {
    datbuf[i] = Wire.read();
  }
}

int ReadDistance() {
  unsigned char retries = 0;
  int distance = -1;

  while (retries < maxRetries) {
    SensorRead(0x00, i2c_rx_buf, 2);
    lenth_val = (i2c_rx_buf[0] << 8) | i2c_rx_buf[1];
    if (lenth_val > 0) { // Valid reading
      distance = lenth_val;
      break;
    }
    retries++;
    delay(10); // Retry delay
  }

  if (distance < 0) {
    Serial.println("Sensor read failed!");
    distance = 0; // Default to 0 on failure
  }

  return distance;
}
```

Photo: *Snippets of C++ code*

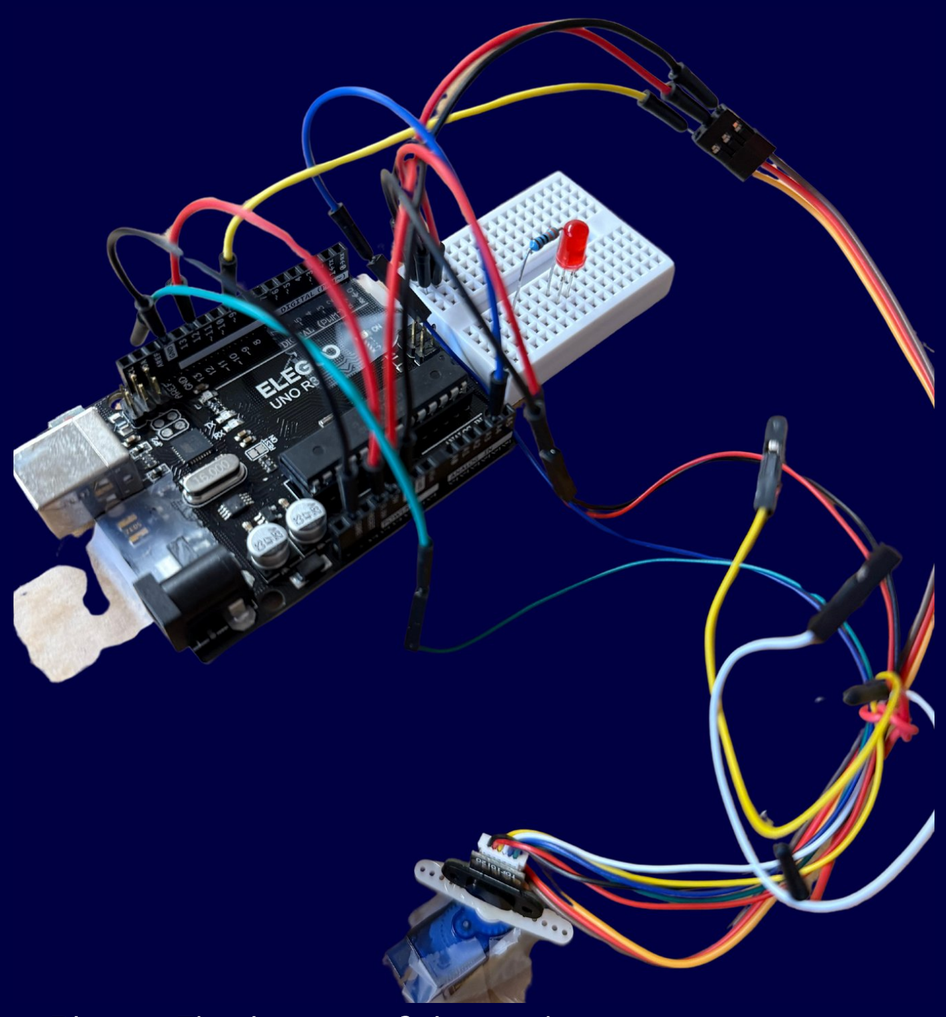


Photo: *Final stage of the project*

Future Improvements

- Enhancements could make the project more practical for real-world applications.
- Sensor data could be tracked and charted on a computer, similar to radar functionality.
- A graphical interface could display distance, direction, and velocity of moving objects within the field.

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